MIL-P-63397A (AR)
11 January 1985
SUPERSEDING
MIL-P-63397 (AR)
12 August 1980

MILITARY SPECIFICATION

PROJECTILE, 155MM: INERT PXR 6325 (M549 SIM) METAL PARTS, ASSEMBLY

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication and assembling of metal parts for Projectile, 155MM: Inert PXR 6325 (M549 SIM)

2. APPLICABLE DOCUMENTS

2.1 The following documents of issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-2550	- Ammunition, General Specifications For
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-S-10520	- Steel, Forging: For Projectile Stock
MIL-S-11310	- Steel Bars, Carbon, Hot Rolled, In Cold Shaping Including Cold Extrusion
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

Federal Test Method

Std No. 141 - Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing
Std No. 151 - Metal, Test Method

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1168 - Lot Numbering of Ammunition

2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)
9236061 - Metal Parts Assy for, 155MM, Inert PXR 6325 (M549 Simulator)
9236062 - Body
9236063 - Projectile, 155MM: Inert PXR 6325 (M549 SIM)
9280373 - Magnetic Particle Standard For Projectile Body, 155MM, Inert PXR 6325 (M549 SIM)
9362569 - Packing & Marking for Pallet, Wood Ammo, for PJCTL 155MM

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASTM A711 - Carbon Steel Blooms, Billets and Slabs for Forging
ASTM E8 - Tension Testing of Metallic Materials
ASTM E10 - Brinell Hardness of Metallic Materials
ASTM E103 - Rapid Indentation Hardness Testing of Metallic Materials

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(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

American National Standards Institute (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Steel quality. The item contractor shall establish procedures to provide a forge quality steel when procuring steel in accordance with ASTM A711, i.e., the steel shall be free of pipe, porosity and heavy segregation. The manufacturer will provide a certification to cite the specific revisions of the industry standard under which the steel was manufactured. Material procured in accordance with MIL-S-10520 or MIL-S-11310 shall comply with the requirements of the applicable specifications.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.3.1 Hot forge heat treat method. Prior to finish machining, projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (^{o}F), followed by an oil quench. After quenching, the body forgings shall be tempered without delay at a temperature above 750 degrees Fahrenheit, suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees Fahrenheit, after which cooling may be accelerated by water spray.

3.3.2 Cold shaping and hot cup - cold draw methods. Cold working shall be so accomplished as to produce in the projectile body the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700 degrees

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Fahrenheit, minimum, for a minimum of 30 minutes following the final forming operation during which time plastic flow takes place.

3.4 Mechanical properties - body. Mechanical properties specified on the applicable drawing shall apply throughout the component.

3.5 Rotating band.

3.5.1 Minimum base metal penetration. There shall be penetration of the base metal at the weld interface in the form of arc cratering or intergranular penetrants such that a proper metallic bond is achieved between the copper and base metal.

3.5.2 Maximum base metal penetration. There shall be no evidence of arc cratering, except for where the starting weld pass is made, when the base metal is machined to 5.950 inches diameter minimum. There shall be no evidence of arc cratering at any location on the base metal when it is machined to 5.850 inches diameter minimum.

3.5.3 Chemical requirements. The iron content of the rotating band shall not exceed 3.0 percent.

3.5.4 Cracks. There shall be no evidence of cracks in the finished machined rotating band. In addition, there shall be no evidence of cracks in the base metal beneath the rotating band at the diameters specified in 4.4.3.3.3.2 and 4.4.3.3.3.3 except that at the 5.910 inch diameter a maximum of seven (7) indications are allowed. No single indication shall be greater than 1/4 inch in length. All indications including those in the starting weld zone must be removed when machined to the 5.850 inch diameters.

3.6 Metal defects. The body shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, slabs and other metal defects.

3.7 Proving ground. The projectile shall show no loss or break up of metal components in the gun bore or in flight.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

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3.9.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.9.2 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.9.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

3.9.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of MIL-A-2550.

3.9.5 Burr. No part shall have a burr which might interfere with the assembly of the round or which might be injurious to personnel handling the item.

3.9.6 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.9 7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

3.9.8 Weld splatter. Copper band weld splatter permissible on exterior surface of projectile provided existing dimensions, tolerances, and requirements are met prior to and after welding. Visual standards will be established for weld splatter.

3.10 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Body	9236062	5
Metal Parts Assy for, 155MM, Inert PXR 6325 (M549S1M)	9236061	5
Projectile, 155MM: Inert PXR 6325 (M549S1M)	9236063	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. Visual standards for examination of band seat after chemical removal of rotating band shall be prepared from the first article sample for acceptable visible intergranular penetration and proper distribution of arc cratering.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assemblies and Components	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body				
	(Dwg. 9236062)				
	Examination for Defects	5 (a)	-	3.1	4.4.2.1, 4.4.2.2
	Mechanical properties - body	5 (a)	-	3.4	4.4.3.2
	Metal Parts Assy for, 155MM; Inert PXR 6325 (M549 SIM)				
	(Dwg. 9236061)				
	Examination for Defects	5 (b)	-	3.1	4.4.2.3
	Rotating band	2 (b)	-	3.5	4.4.3.3
	Salt spray	3 (b)	-	3.1	4.4.3.4.1
	Projectile, 155MM; Inert PXR 6325 (M549 SIM)				
	(Dwg. 9236063)				
	Proving ground test	10	-	3.7	4.5.4

NOTES:

(a) Use same five (5) samples for these tests and examinations.

(b) Use same samples for these tests and examinations.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, the following applies:

a. Heat treatment lots. Heat treatment lots shall contain heat treated bodies from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

b. Stress relieved lots. Stress relieved lots shall contain stress relieved bodies, from not more than one heat of steel, that have been cold shaped by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

c. Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either (a) or (b) above, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

d. Projectile lots. When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

e. Shipping lots. Shipping lots shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawings, revision, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lots thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procurring contracting officer.

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4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63397A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body, Prior to Banding	1 OF 1	9236062
			NEXT HIGHER ASSEMBLY: 9236061

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Diameter of rotating band seat		0.40%	3.1	Gage
102	Circular runout of rotating band seat diameter		0.40%	3.1	Gage
Minor					
201	Surface finish of rotating band seat		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63397A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.2	Body		9236062
			NEXT HIGHER ASSEMBLY
			9236061

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Perpendicularity of rear face with body O.D.		0.40%	3.1	Gage
102	Length, total		0.40%	3.1	Gage
103	Thickness through base		0.40%	3.1	Gage
104	Circular runout of thread pitch diameter with body and front face		0.40%	3.1	Gage
105	Circular runout of ogive with datum -A-		0.40%	3.1	Gage
106	Circular runout of bourrelet with datum -A-		0.40%	3.1	Gage
107	Circular runout of rotating band with datum -A-		0.40%	3.1	Gage
108	Circular runout of body O.D. in front of boattail with datum -A-		0.40%	3.1	Gage
109	Circular runout of boattail with datum -A-		0.40%	3.1	Gage
110	Diameter of bourrelet, min		0.40%	3.1	Gage
111	Diameter of body O.D. Datum -A-		0.40%	3.1	Gage
112	Diameter of body in front of boattail		0.40%	3.1	Gage
113	Diameter of rotating band O.D.		0.40%	3.1	Gage
114	Length from front face to forward basic diameter		0.40%	3.1	Gage
115	Length from front face to rear basic diameter		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9236062
4.4.2.2	Body (continued)				NEXT HIGHER ASSEMBLY 9236061
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Location of rotating band from rear of boattail		0.40%	3.1	Gage
117	Thickness of body wall, min.		0.40%	3.1	Gage
118	Variation in wall thickness at ogive and body		0.40%	3.1	Gage
119	Depth of undercut forward of rotating band, max.		0.40%	3.1	Gage
120	Pitch diameter of threads		0.40%	3.1	Gage
121	Minor diameter of thread, max.		0.40%	3.1	Gage
122	Length of effective thread, min.		0.40%	3.1	Gage
123	Length of thread stock, max.		0.40%	3.1	Gage
124	Properties of rotating band		Note a	3.5	4.4.3.3, 4.5.5
125	Mechanical properties of body		Note b	3.4	4.4.3.2
Minor					
201	Diameter of thread chamfer		0.65%	3.1	Gage
202	Angle of thread chamfer		0.65%	3.1	Gage
203	Diameter of rotating band groove		0.65%	3.1	Gage
204	Diameter of rear of boattail		0.65%	3.1	Gage
205	Length from front face to rear of bourrelet		0.65%	3.1	Gage
206	Width of undercut forward of rotating band, max.		0.65%	3.1	Gage
207	Profile of rotating band		0.65%	3.1	Gage
208	Angle at front of rotating band		0.65%	3.1	Gage
209	Angle at rear of rotating band relief groove		0.65%	3.1	Gage

NOTES:

a. See paragraph 4.4.3.3

b. See paragraph 4.4.3.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9236062
4.4.2.2	Body (continued)				NEXT HIGHER ASSEMBLY 9236061
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210	Diameter of band seat adjacent to rotating band		0.65%	3.1	Gage
211	Circular Runout of band seat diameter adjacent to rotating band		0.65%	3.1	Gage
212	Depth of undercut rear of rotating band (if applicable)		0.65%	3.1	Visual
213	Width of undercut rear of rotating band (if applicable)		0.65%	3.1	Visual
214	Surface finish improper (rotating band)		0.65%	3.1	Visual
215	Surface finish improper (entire body except rotating band)		0.65%	3.1	Visual
216	Protective finish improper or inadequate		0.65%	3.1	Visual
217	Anti-seize compound missing from threads		0.65%	3.1	Visual
218	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9236061
4.4.2.3	Metal Parts Assy for 155MM, Inert, PXR 6325 (M549 Simulator)				NEXT HIGHER ASSEMBLY 9236063
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Diameter of bourrelet, max. (after paint)		100%	3.1	Gage
102	Rotating band damaged		0.45%	3.1	Visual
103	Salt spray test		Note a	3.1	4.4.3.4.1, 4.5.3
Minor					
201	Foreign matter in cavity		0.65%	3.1	Visual
202	Paint on threads		0.65%	3.1	Visual
203	Paint on rotating band bearing surface		0.65%	3.1	Visual
204	Protective coating inadequate or damaged		0.65%	3.1	Visual
205	Anti-seize compound missing from threads		0.65%	3.1	Visual
206	Marking missing, incorrect, or illegible		0.65%	3.1	Visual
207	Grommet missing		0.65%	3.1	Visual
208	Shock attenuating plug missing		0.65%	3.1	Visual
209	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

a. See Paragraph 4.4.3.4.1

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.4	Pallet		1 OF 1	9362569
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Pallet damaged or improperly assembled		0.40%	3.1	Visual
102	Projectile of different zones on the same pallet		0.40%	3.1	Visual
103	Strapping or staple missing or loose		0.40%	3.1	Manual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.1	Visual
202	Marking misleading or unidentifiable		0.65%	3.1	Visual
203	Strapping or staple improperly engaged		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Steel quality. Records of test documentation shall be maintained and made available for Government review to provide evidence of compliance with the requirements of 3.2.

4.4.3.2 Mechanical properties.

4.4.3.2.1 Hardness.

4.4.3.2.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.4.1a) after heat treatment, but prior to final machining of the bourrelet diameter, shall be subjected to a hardness test with a Brinnel Hardness Tester or any other method approved by the procuring activity. Hardness impressions shall be made on the body bourrelet diameter and used as the basis for selecting the tension test samples of paragraph 4.4.3.2.2.1.

4.4.3.2.1.2 Stress relieved lots. Twenty-five (25) cold shaped bodies from the first batch of each stress relieved lot (as defined by 4.4.1b) processed through the stress relief furnace and 5 bodies from each one hours' production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product.

4.4.3.2.1.3 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.2.

4.4.3.2.2 Tension test

4.4.3.2.2.1 Heat treatment lots. Two (2) projectile bodies, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

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4.4.3.2.2.2 Stress relieved lots. Two (2) projectile bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five projectiles shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five (25) degrees Fahrenheit but not below the minimum temperature of 700 degrees Fahrenheit and the minimum 30 minutes soak time. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each eight (8) hours by an independent method.

4.4.3.2.2.3 Miscellaneous lots. Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.4.3.2.2.1 or 4.4.3.2.2.2, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawings shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.4.3.3 Rotating band. A sample shall be taken, for chemical analysis and minimum penetration, from each overlay band machine, from each eight (8) hour shift or fraction of a shift from machines in use. The randomly selected projectile bodies to be used as samples, and all bodies they represent, shall be identified for each machine and shift until all tests and examinations of 4.4.3.3 are complete. The bodies selected are not required to be heat treated.

4.4.3.3.1 Chemical analysis. The iron content of the copper shall be determined by chemical analysis of chips machined from the copper overlay. The chips for the sample shall consist of copper overlay between 6.212-.005 and 6.162-.005 inches diameter. All of the chips shall be collected and crushed, with a sample portion of 20 $\pm$ 5 grams then randomly selected and dissolved. Chemical analysis shall be conducted in accordance with Federal Test Method Standard No. 151 except that the sample shall not be cleaned with a magnet. Samples having iron content of 2.0 percent max. shall be considered acceptable. Any sample having iron content 2.0 to 3.0 percent max. shall be considered acceptable upon demonstration of compliance, for that sample's base metal, with the requirements of 3.5.2 and 3.5.4 (per 4.4.3.3.3). Any sample having iron content in excess of 3.0 percent, and that portion of production represented by that sample, shall be considered rejected.

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4.4.3.3.2 Minimum penetration. Each sample body selected in accordance with 4.4.3.3 shall be inspected for minimum base metal penetration. The remanent rotating band shall be dissolved by any method which will not damage the weld interface surface. The remanent band may be machined to 6.040 inches minimum diameter prior to dissolving the remaining copper. The surface shall be visually observed for minimum penetration as follows:

a. The sample shall be considered acceptable if the quantity and distribution of arc cratering equal or exceed that of the appropriate visual standards.

b. The sample shall be considered acceptable if the visible intergranular penetrants, under magnetic particle inspection, equal or exceed those of the appropriate magnetic particle standard.

c. The sample shall be considered acceptable if all of the following conditions are satisfied:

1) No uncratered area exceeds one square inch. Evaluate utilizing an overlay with a 1 x 1 inch square. Overlay shall be positioned with top and bottom edges of square parallel to the edges of the band seat.

2) The sum of uncratered areas does not exceed six (6) square inches. Evaluate utilizing an overlay divided into 1/4 x 1/4 inch (1/16 square inch) boxes. Overlay shall be positioned with top and bottom edges of square parallel to the edges of the band seat. All uncratered areas which exceed 1/2 x 1/2 or 1 x 1/4 inch shall be totaled to the nearest 1/16 square inch. The overlay shall be of sufficient size to cover the entire band seat and shall not be moved during evaluation of surface for items "2", "3" and "4". One uncratered area, within 1 inch of the weld stop-start, may be disregarded if the width is less than 1/2 inch.

3) Uncratered areas are separated by 1/2 inch minimum. Evaluate utilizing overlay as in 2) above.

4) No uncratered area is within 1/4 inch of the leading or trailing edge of the band seat. Evaluate utilizing overlay as in 2) above, and disregarding one area within 1 inch of the weld stop-start.

5) Each uncratered area, when examined by magnetic particle inspection, exhibits four (4) or more intergranular penetrants. Uncratered areas, identified in 2) above, shall be examined by magnetic particle inspection for evidence of 4 penetrants per area. One uncratered area, within 1 inch of the weld stop-start, may be disregarded if the width is less than 1/2 inch.

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Any sample failing to meet the criteria of a), b) or c) above, and that portion of production represented by that sample, shall be considered rejected.

4.4.3.3.3 Maximum penetration and cracks. Every third day's sample, from each band machine selected in accordance with 4.4.3.3 shall be examined for maximum penetration and cracks. Any sample failing to meet the applicable criteria, when examined in accordance with 4.4.3.3.3.1, 4.4.3.3.3.2 and 4.4.3.3.3.3, and that portion of production represented by that sample, shall be considered rejected. If the third day's sample is rejected, the second day's sample may be examined for acceptance of the two day's production, from that machine which the sample represents. If the second day's sample, and production represented by that sample, is rejected, the first day's sample may be examined for acceptance of that day's production.

4.4.3.3.3.1 Maximum penetration (other than starting weld pass). The sample band seat shall be machined to a minimun diameter of 5.950 inches. The band surface shall be visually examined for presence of arc cratering, except for the area at the start of the weld pass. The starting weld pass area is defined as 1/2 inch maximum on each side of the start-up weld center line.

4.4.3.3.3.2 Cracks. Upon completion of 4.4.3.3.3.1, the band seat shall be machined to a minimum diameter of 5.910 inches. The entire band seat area shall be examined, by magnetic particle inspection, for cracks and intergrannular penetrants. The presence of indications in the starting weld pass area shall not be cause for rejection. If the entire band seat area is free of indications under magnetic particle inspection, and free of arc craters under visual examination, the sample shall be considered final accepted and examination under 4.4.3.3.3.3 is not required.

4.4.3.3.3.3 Maximum penetration and cracks (including starting weld pass). Upon completion of 4.4.3.3.3.2, and if required, the band seat shall be machined to a minimum diameter of 5.850 inches. The entire band seat area shall be visually examined for the presence of arc cratering, and examined by magnetic particle inspection for cracks and intergrannular penetrants. Presence of any craters, cracks or penetrants shall be cause for rejection of the sample and all production represented by the sample.

4.4.3.4 Cleaning and application of protective coating. The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements of TT-C-490 shall be cause for taking the corrective action recommended therein.

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4.4.3.4.1 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.3.5 Proving ground test. Ten (10) body assemblies from each production lot shall be subjected to this test. If any of the sample body assemblies fail to meet the requirements of 3.7, the lot shall be rejected. The test shall be conducted as specified in 4.5.4.

4.4.3.6 Tool control dimensions.

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs.), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted providing the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of the tool control dimension(s) will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the table starting in Paragraph 4.4.2.1. The contractor shall submit for approval of inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Mechanical properties.

4.5.1.1 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

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a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone.

b. Hardness tester periodic verification during each operating shift shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The Hardness Tester shall use a standard 10 MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 300Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

4.5.1.2 Tension test. From the samples provided by the heat treat or stress relief lots, two tension specimens conforming to the largest obtainable standard round type specimens, shall be tested in accordance with the procedure prescribed by ASTM E-8 and shall be taken longitudinally from the opposite side of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their locations are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Present elongation shall be recorded to the nearest tenth (0.0 percent).

4.5.2 Protective coating. The procedure for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification.

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4.5.3 Salt spray. The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.4.1.

4.5.4 Proving ground functional tests. The tests shall be performed at a Government Proving Ground in accordance with the applicable Acceptance Test Procedure. The sample body assemblies selected in accordance with 4.4.3.5 shall be loaded and assembled as specified on Dwg. 9236063 and fired at ambient temperature in a M185 or M199 series Cannon using a PXR-6297 proof charge. The propelling charges shall be uniformly temperature conditioned to +145°F minus 5°F for a minimum of 12 hours prior to the test. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile to assure compliance with 3.7. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressures will be recorded for information only.

4.5.5 Magnetic particle inspection. The inspection shall be in accordance with specification MIL-I-6868 using the continuous, wet, fluorescent method. The copper welded overlay band seat shall be inspected after inducing a circular magnetic field and again after inducing a longitudinal magnetic field. Acceptance criteria shall be in accordance with 4.4.3.3.2 or 4.4.3.3.3 for indications at the 5.910 inch diameter and 5.850 inch diameter and for maximum base metal cratering at the 5.950 inch diameter. Only the machined area beneath the rotating band must be covered with suspension. The magnetic particle content of the suspension shall be maintained at the level recommended by the suspension manufacturer. After draining the excess suspension, the band seat shall be inspected under ultraviolet light. The adequacy of the inspection process shall be checked at the start of daily production and at eight hour intervals thereafter using a standard in accordance with dwg. 9280373. The summation of the seam definition between the plug and body on the outside surface after circular and longitudinal magnetization shall not be less than the entire 360° circumference of the plug. The magnetizing current shall be adjusted to meet this requirement. If the inspection process is determined to be inadequate the condition shall be corrected and all bodies accepted since the last standard check shall be reinspected. The requirement of specification MIL-I-6868 to check system effectiveness by means of the Ketos steel ring specimen shall be waived.

4.5.5.1 Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Leval I. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

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5. PACKAGING

5.1 Packing. Level A. The projectile shall be packed in accordance with dwg. 9362569.

5.2 Marking. Marking shall be in accordance with dwg. 9362569.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for practice usage for use with the Fuze, and fired from the 155MM Howitzer.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and/or equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Munitions and Chemical Center, ATTN: DRSMC-QAR-I(D), Dover, New Jersey 07801.

6.4 It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions and mechanical properties and metal inegrity specified as an end item requirement.

6.5 The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade selected by the projectile manufacturer and complying with specification ASTM A711.

6.6 In the application of carbon steel, blooms, billets, slabs and bars forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequecne of operations, tooling designs, handling devices, and processing techniques having capabilities of compensating for such variables.

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6.7 Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and/or metallurgical properties of the metal involved.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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